# Speed Modelling Guide Book

# 1. Introduction

The Speed Modelling Event is a CAD modelling competition that tests the participant's speed, accuracy, and modelling skills under limited time conditions.

In this event, participants will be given a 2D sketch of a component along with the material specification through the event website. Using the given information, participants must create the 3D model of the component in the permitted CAD software, assign the specified material, and calculate the weight of the model.

After completing the model, participants must upload the required files and the calculated weight through the website before the allotted time ends.

The evaluation and leaderboard updates will be handled automatically by the website based on the submission time and accuracy of the submitted result.

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# 2. Permitted Software

Participants are allowed to use any one of the following software:

- Autodesk Fusion
- CATIA V5

No other CAD software will be permitted during the competition.

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# 3. System Requirements

Participants may either:

- Use their own laptop, or
- Use the system provided by the organizers (subject to availability).

Participants using personal laptops are responsible for ensuring that:

- The required software is installed properly
- The system is functioning correctly

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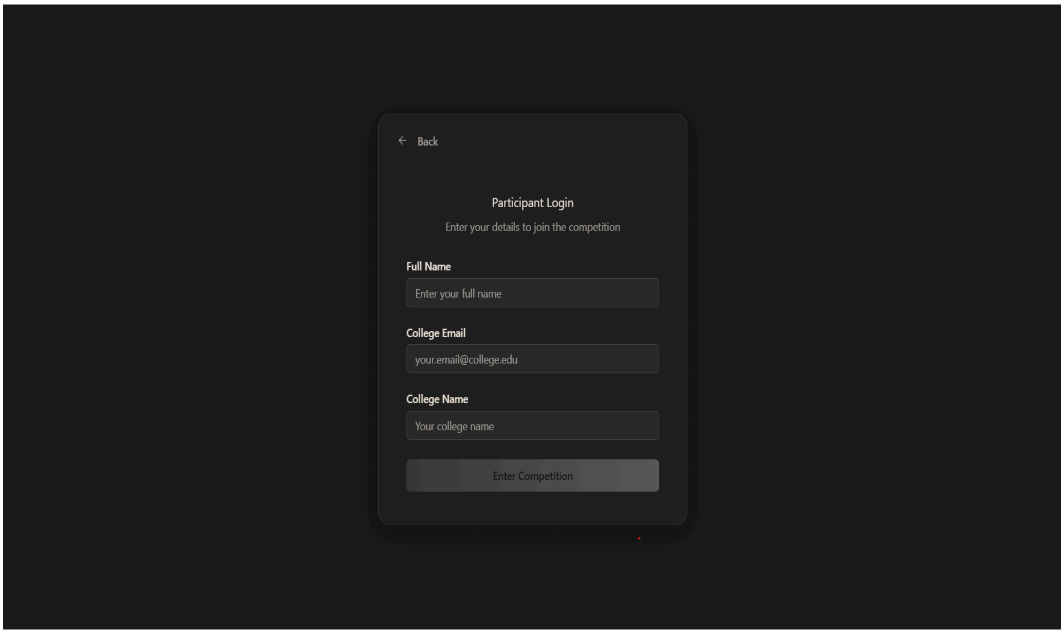
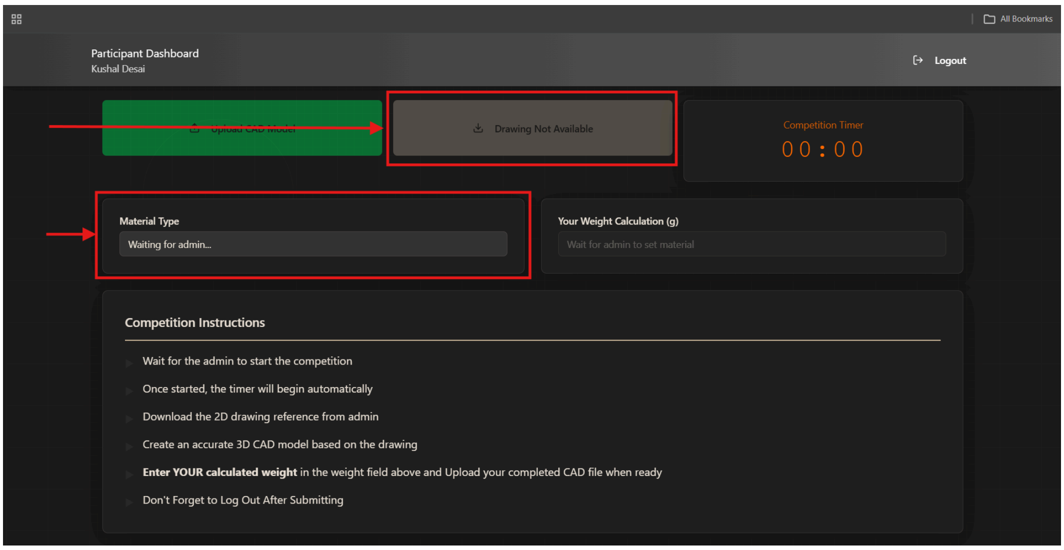
# 4. Competition Workflow & Submission Guide

## Step 1: Login to the Website

- Open the event website and log in using your college email ID.
- After logging in, you will be redirected to the participant dashboard where all competition activities will take place.




## Step 2: Download the 2D Sketch and Note the Material

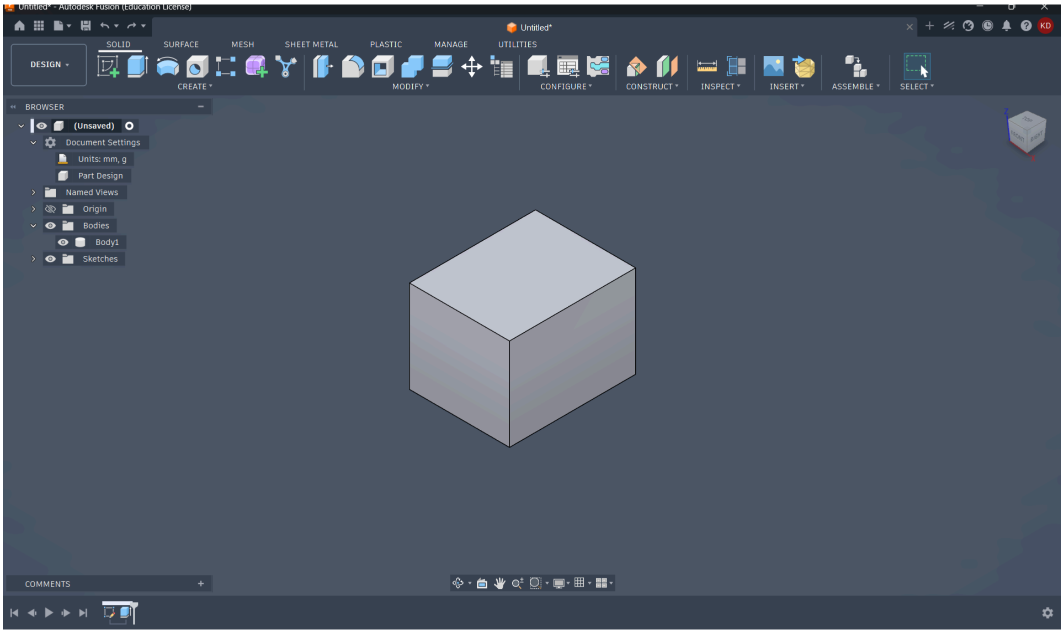



- Once the competition starts, the 2D sketch will be available on the dashboard.

- Download the sketch and carefully check all the dimensions and features before starting the model.
- The material specified for the component will also be displayed on the website. Make sure to note it down, as the same material must be applied to the model later for weight calculation

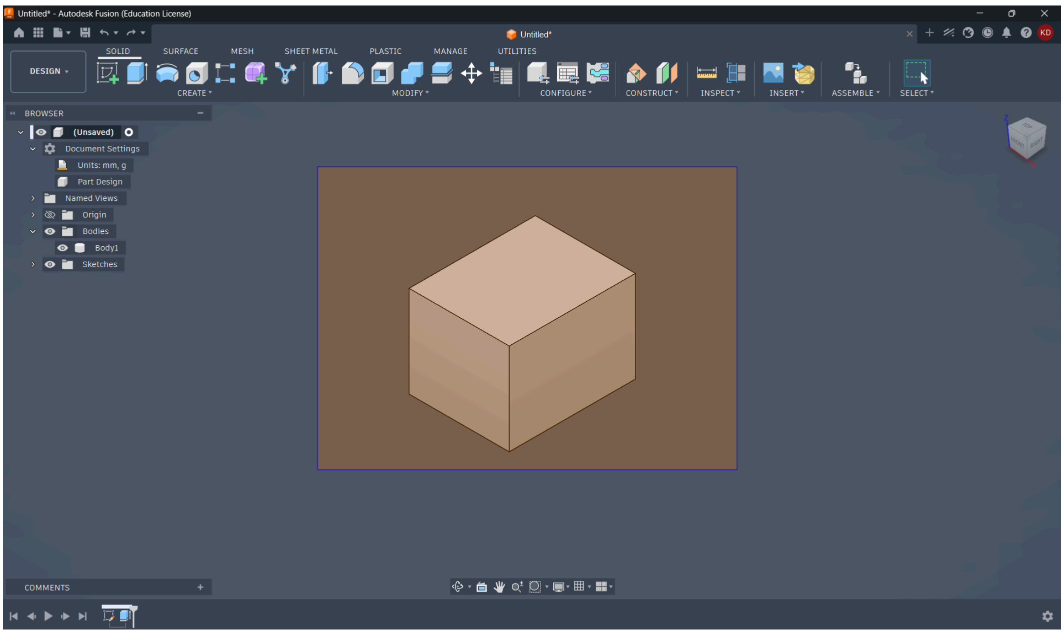
## Step 3: Create the 3D Model

- Open Autodesk Fusion and start creating the 3D model based on the given sketch.
- Ensure that all dimensions and features are modelled accurately according to the drawing.

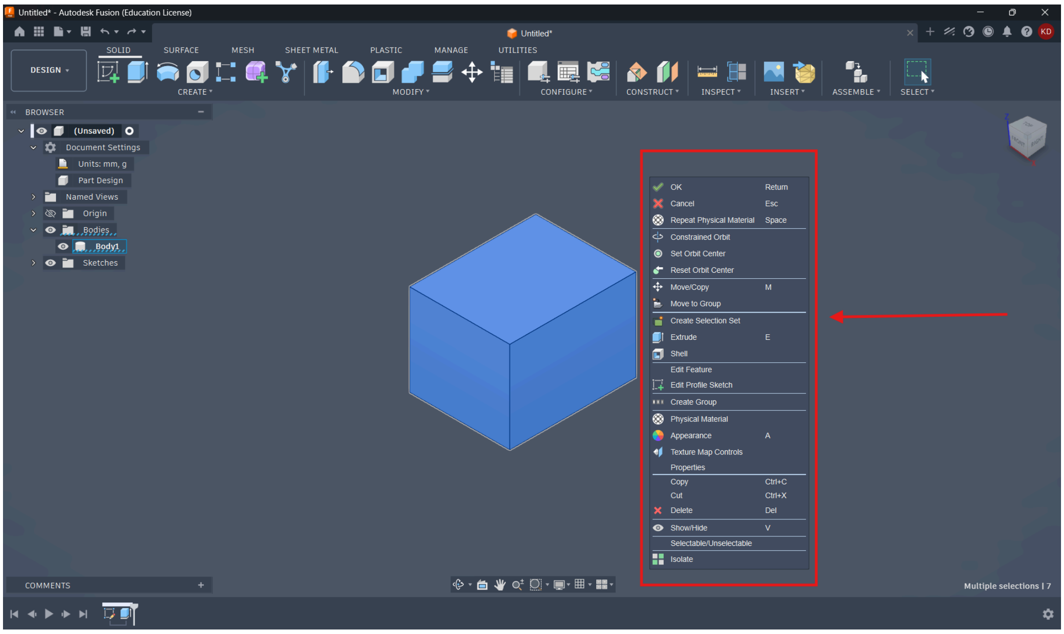



## Step 4: Apply the Specified Material

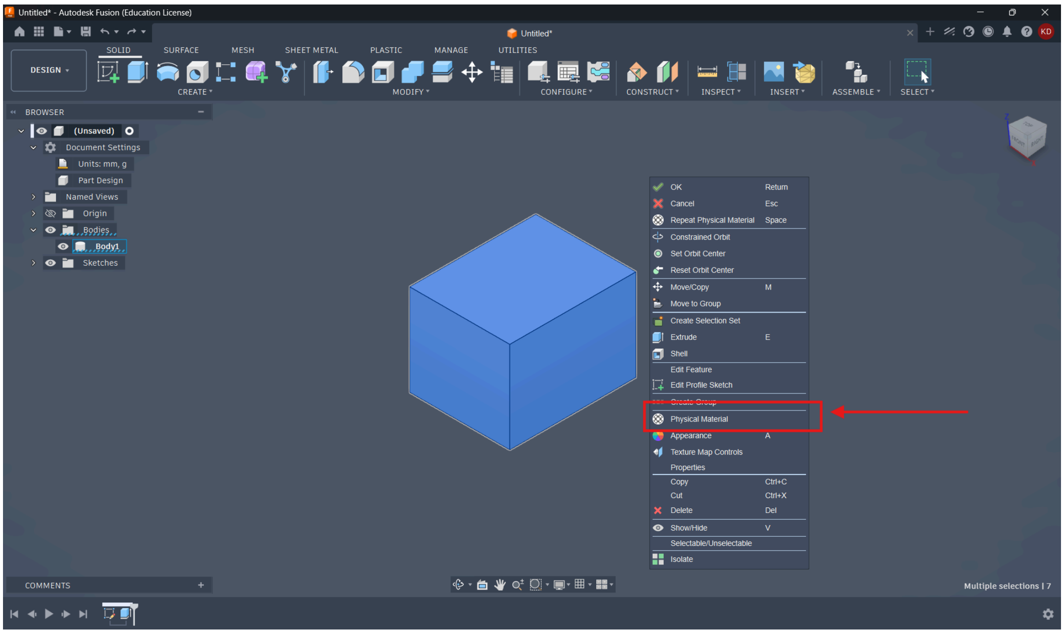
- After completing the 3D model, select the complete model/body by left-clicking and dragging the cursor across the model to create a selection box.



- While creating the selection box, make sure that the complete 3D model is fully covered within the selected area and no feature or portion of the model remains outside the selection.
- After confirming that the entire model has been selected properly, right-click on the selected model to open the options menu.

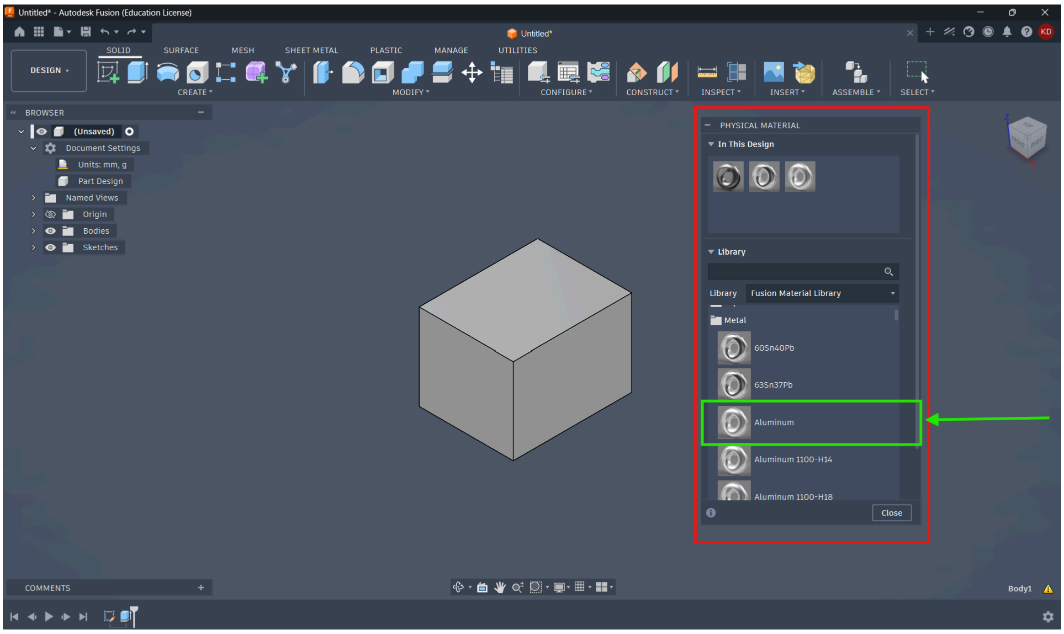



- From the dialogue box, click on **Physical Material**.



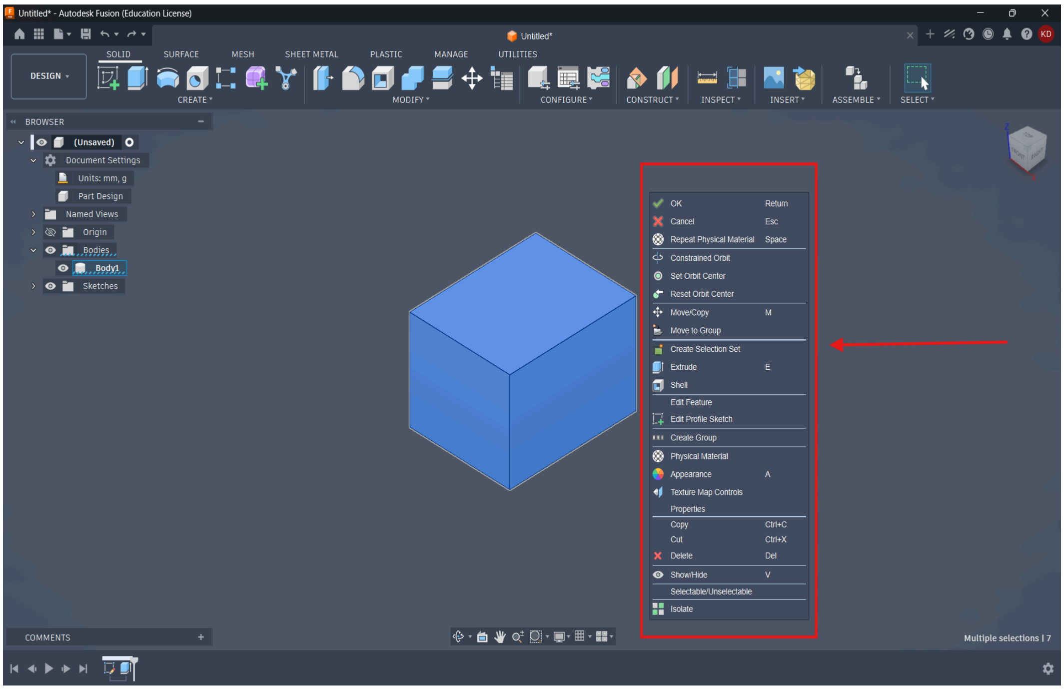

- After selecting the **Physical Material** option, the Fusion material library window will open on the screen.
- The library contains different categories of engineering materials available in Fusion.
- Locate the material specified on the competition website carefully from the material library.
- Once the correct material is found, click and drag the material onto the selected 3D model/body.

- After applying the material successfully, the physical properties of the model will update accordingly.

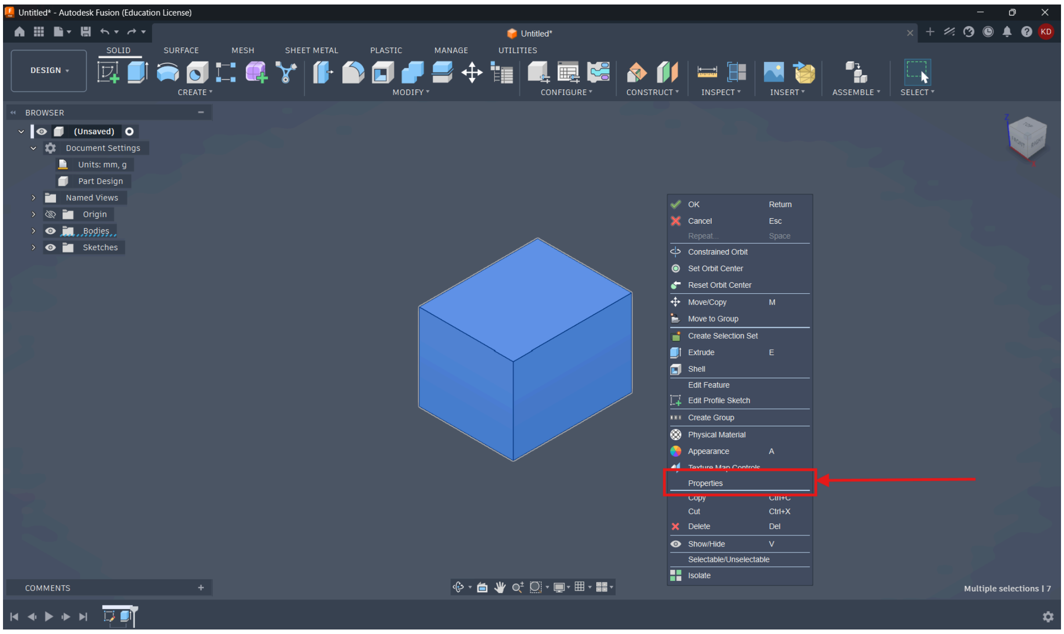



## Step 5: Obtain the Weight/Mass of the Model

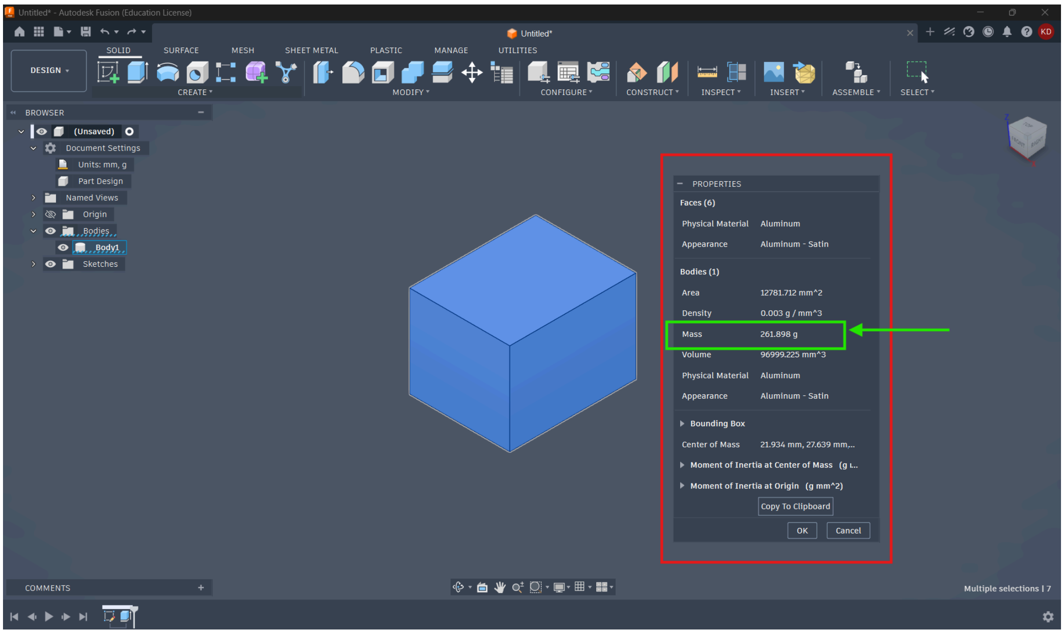
- Select the complete 3D model/body by left-clicking and dragging the cursor across the model to create a selection box.
- Ensure that the entire model is completely selected before proceeding.
- Once the model is selected, right-click to open the options menu.



- From the dialogue box, click on **Properties**.



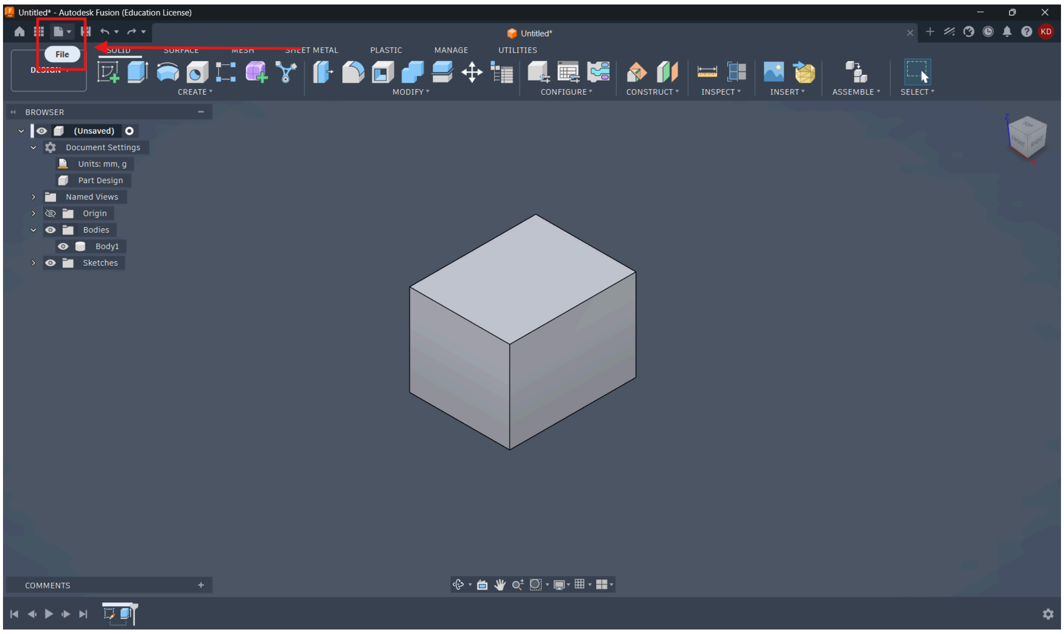

- A properties window will open showing various physical properties of the model, including the weight/mass value.
- Note down the displayed weight/mass carefully before submission.



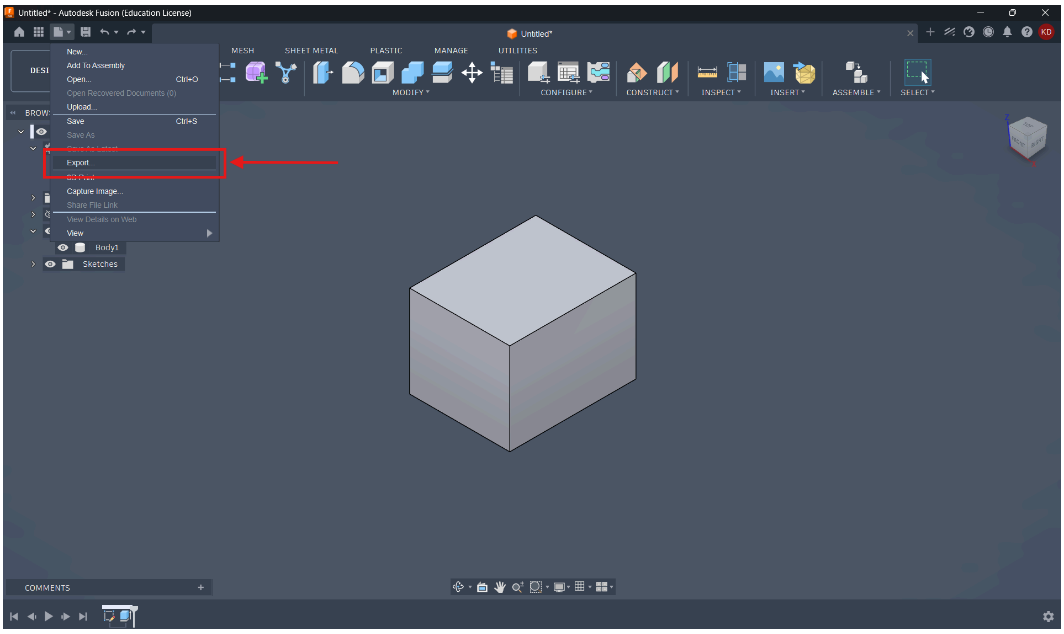

- Verify that the material has been applied correctly before recording the weight/mass value.

# Step 6: Export the Model in STEP Format

- After verifying the model and weight/mass, export the completed model in STEP format.
- In Fusion, click on the **File** option located at the top-left corner of the screen.

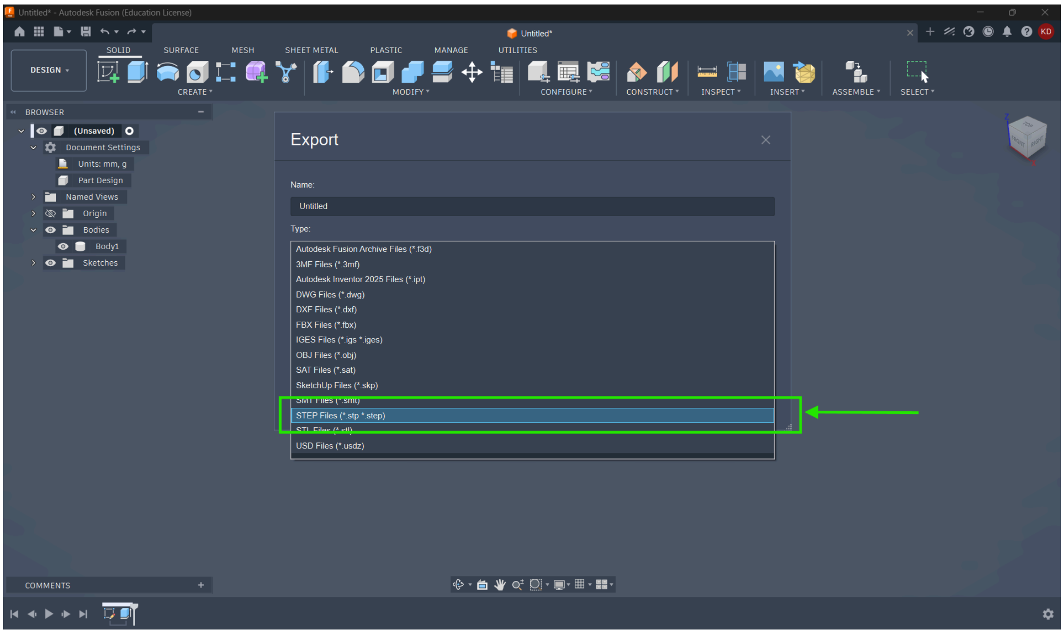



- From the File menu, click on **Export**.




- After clicking on the Export option, the export window will open on the screen.
- Different file format options will be available in the export settings.
- From the available file types, select the STEP export format.
- Ensure that the selected file type is **STEP (.stp)** before exporting the model.

**STEP (.stp)**




- Choose the save location and export the file to your system

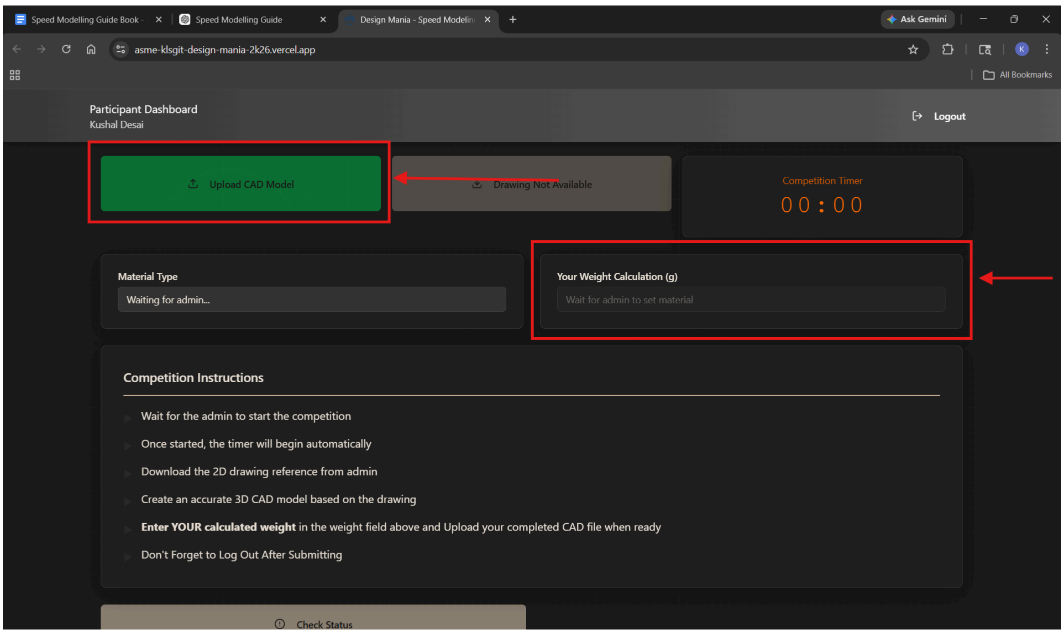
### File Naming Format

- Name the STEP file using your participant name before uploading.

**Example:**
`Kushal.stp`

## Step 7: Upload Your Submission

- Return to the competition website after exporting the STEP file.
- Upload the exported STEP (.stp) file in the upload section provided on the website.
- Enter the calculated weight/mass value obtained from Fusion in the specified input field.



### Step 8: Final Submission

- After verifying all details, click on the **Submit** button on the website to complete your submission.
- Once submitted successfully, the website will automatically record your submission time.
- After the competition timer ends, further submissions will not be accepted by the website.

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# 5. Submission Guidelines

- The CAD file must be submitted only in `.stp` format.
- Participants must ensure that the uploaded files are correct and not corrupted.
- Incomplete submissions may not be considered for evaluation.
- Participants must complete and submit their work within the allotted competition time.
- The timer displayed on the website will be considered as the official event timer.
- Once the time limit ends, the website will stop accepting submissions automatically.
- Participants who fail to submit within the given time will be disqualified.
- All submitted files must be named according to the participant’s name.

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# 7. Rules and Regulations

- Only individual participation is allowed unless stated otherwise by the organizers.
- Participants must use only the permitted CAD software.
- The submitted model should match the provided 2D sketch accurately.
- The specified material must be applied correctly before calculating the weight.
- Any form of malpractice or unfair means will lead to disqualification.
- Use of unauthorized software or external assistance is strictly prohibited.

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# 8. Evaluation Criteria

The evaluation will be carried out automatically through the website based on:

- Accuracy of the submitted weight/mass
- Time taken for submission

Participants with more accurate results and lower completion time will secure higher positions on the leaderboard.

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# 9. Disqualification

A participant may be disqualified if:

- The submission is made after the allotted time
- Incorrect or incomplete files are uploaded
- Unauthorized software is used
- Any unfair means or malpractice is identified

The decision of the organizers regarding the competition and results will be final.

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# 10. Pre-Competition Recommendations

Participants are advised to:

- Practice CAD modelling and STEP export procedures beforehand
- Participants are advised to learn how to properly apply materials and obtain the weight/mass properties of a model in their selected CAD software before the competition.
- Verify software functionality before the event
- Manage time carefully during the competition

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## Note:

**The 3D model, dimensions, material, and weight/mass values shown in the images are intended solely for demonstration and explanatory purposes. Autodesk Fusion has been used in this guide to illustrate the workflow and submission procedure; however, participants may use either Autodesk Fusion or CATIA V5 during the competition. The actual component, specified material, and calculated values used in the event may vary.**